

Jacobian-lens trajectories under medical SFT

`gemma-3-270m-it` and `gemma-3-1b-it`

July 28, 2026

Full fine-tune on `mix_default`, lr $1e-5$, log-spaced checkpoints to step 4882

Setup

- **Two arms**, identical seed and data order: `gemma-3-270m-it` and `gemma-3-1b-it`, full FT, lr $1e-5$, completion-only loss.
- **Log-spaced checkpoints** $\{1, 2, 4, \dots, 4096\}$ + final (4882). $t = 0$ is the untouched base, prepended by the probe.
- **Lens refit per checkpoint** on a frozen 1000-sequence generic (non-medical) WikiText corpus, taken from `jlens`'s own `load_wikitext_prompts` — the lens is a function of the weights.
- **Concordance** on the frozen 193-prompt probe set: 50 `general` + 50 `medical` (hand-authored one-hop cloze, style-matched control pairs) + 93 `multihop` from `jlens`'s `lens-eval-multihop.json`.
- **Behavioral** trajectory in parallel: MedQA, MedMCQA, PubMedQA, reported as *parsed* accuracy with parse rate alongside.

All lens numbers below are the **mean over the last $n/5$ source layers** (the probe's `kl_final` / `top1_final`): 3 layers at 270m, 5 at 1b. Where the single final layer disagrees, it is called out explicitly — see the read-out caveat. Coverage: 14 points (270m), 15 points (1b) — both complete.

The probe step in detail

Runs in `.venv-jlens` (transformers 5.14.1 + jlens @ 581d398) — a second env, because the training pin 4.53.2 is hard-incompatible. Imports nothing from `gemma_med`.

Per checkpoint (base prepended as step 0, then `checkpoint-*`):

1. **Load** → `jlens.from_hf(hf, tok)`. Forced text-only Gemma3ForCausalLM; any loader leaving a weight missing is *rejected*, not accepted.
2. **Fit** a fresh lens: `jlens.fit(model, corpus, dim_batch=16, max_seq_len=128)` on the frozen 1000-seq WikiText corpus. *Dominates runtime*.
3. **Concordance**: for each of the 193 probes, `lens.apply(model, prompt, positions=[-1])` → per source layer, top-1 agreement and $KL(model \parallel lens)$ at the last token.
4. **Drift** vs. $t=0$, per layer: relative Frobenius $\|J_l - J_l^0\| / \|J_l^0\|$ and cosine.

Output. One `fsync'd` JSON row per checkpoint in `metrics.jsonl`, plus W&B scalars keyed by training step. Resumable: completed steps are skipped and the fit checkpoints every 50 prompts, so a requeue does not wipe the trajectory.

Invariants that make the trajectory comparable

- **One load path for all points.** Letting `Auto*` pick per-path would read $t=0$ through the multimodal wrapper and later points through the text model — two forwards on one trajectory, silently corrupting drift.
- **Lens refit every time.** J is a function of the weights.
- **Corpus and probe set frozen** across all checkpoints and sizes.
- **$t=0$ lens cached in fp32**, so a resumed run's drift baseline matches a fresh run's exactly.

Four findings

1. **Drift has a sharp onset, then saturates.** The Jacobian finishes moving in the first ~ 100 steps; the remaining 97% of training adds nothing.
2. **Medical faithfulness improves at 1b, is flat at 270m — and generic faithfulness degrades at both.** The cost of medical SFT is paid on the out-of-domain cues.
3. **Behavior confirms the split.** 1b gains +5–7 pp parsed accuracy; 270m gains nothing and only churns format.
4. **Drift leads format collapse, both lead accuracy.** The lens finishes moving before the model has learned any medicine.

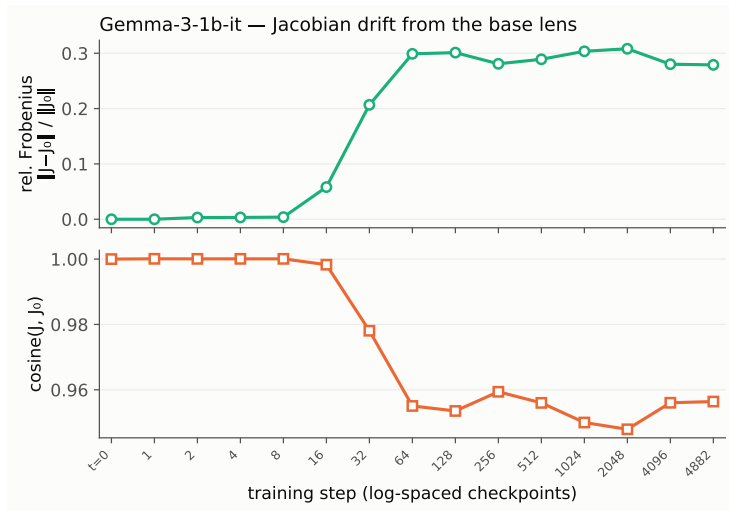
1. Drift onsets sharply, then saturates

	onset	plateau	final relfro / cos (layer mean)
270m-it	steps 32–64	~step 128	0.249 / 0.976
1b-it	steps 8–32	~step 64	0.279 / 0.956

- Nothing moves for the first ~ 8 steps (mean rel. Frobenius ≤ 0.004).
- Then the Jacobian swings hard over roughly *one decade* of steps, and flatlines.
- 1b: 0.299 at step 64 $\rightarrow$ 0.279 at step 4882. The last 4000 steps buy **zero** additional lens movement.

Most robust result — holds at both sizes, and under either the layer mean or the per-layer max (max: 0.349 / 0.398).

1. Drift — 1b-it



2. Medical holds or improves; generic degrades

Lens↔model KL, mean over the last $n/5$ layers (the probe's `kl_final`) — *lower = lens predicts the model better*. Base → final checkpoint.

	KL general			KL medical			KL multihop (OOD)		
270m-it	3.407	→	3.497 (+0.09)	5.388	→	5.340 (−0.05)	1.958	→	1.975 (+0.02)
1b-it	2.684	→	3.229 (+0.54)	4.682	→	4.314 (−0.37)	2.423	→	3.430 (+1.01)
top-1 medical:	270m 0.260 → 0.260 (flat)			1b 0.448 → 0.572					

- **Same direction at both sizes**, differing in magnitude: medical flat-to-better, general and multihop worse.
- **1b moves 6–59× further** on every cue (general 6.0×, medical 7.6×, multihop 59×). At 270m the lens barely registers the fine-tune — consistent with 270m learning no medicine behaviorally.

Read-out caveat: at the *single* final layer, 270m medical reads 1.861 → 2.137 (+0.28, worse) — the one sign flip across all six cue/size pairs. Do not read 270m medical as a degradation: under the layer mean used here it is flat on both KL (−0.05) and top-1 (0.260 → 0.260). At the single final layer its top-1 *also* falls (0.44 → 0.38), so the two read-outs disagree on that cue in both metrics.

2. What is really happening: the domain gap narrows

Within each arm the medical cue starts ~ 2 nats less faithful than general, and both arms narrow that gap — but 1b narrows it **6.6 $\times$ harder**.

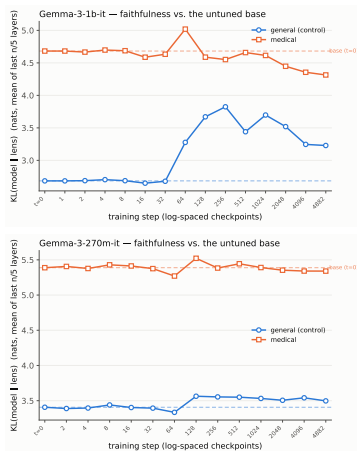
	gap $t=0$	gap final	Δ
270m-it	1.981	1.843	-0.138
1b-it	1.998	1.085	-0.913

Crucially the gap closes **from both ends** — medical improves *and* general regresses toward it. It is not a free gain.

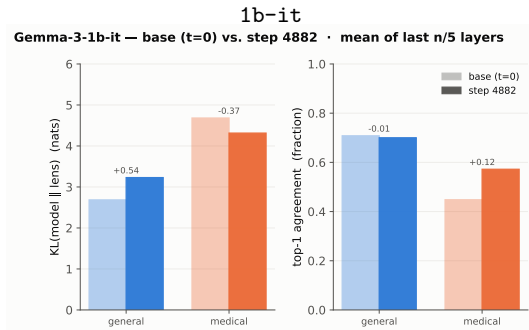
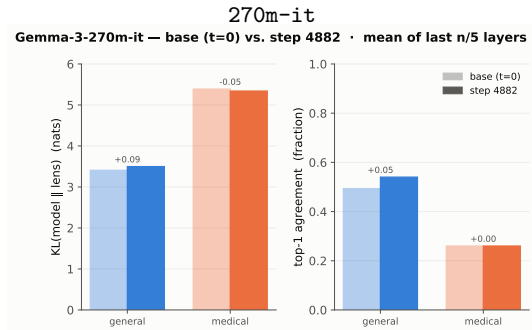
multihop, the held-out externally-sourced cue, is the cleanest read and degrades at *both* sizes (+0.02 / +1.01).

Generic-domain faithfulness is what medical SFT costs you.

The $t=0$ gaps landing 0.017 nats apart is a coincidence, not a finding — absolute KL is not comparable across tokenizers and hidden dims. The claim is the within-arm



2. Base vs. finetuned, side by side



Pale bar = untuned base ($t=0$), solid = step 4882. Both figures are pinned to the same axes (KL 0–6 nats, top-1 0–1), so the panels are directly superimposable — but compare the Δ labels, not the bar heights: absolute KL is not comparable across sizes (different tokenizers and hidden dims). Read that way, 270m's largest KL move (+0.09) is about a quarter of 1b's smallest (−0.37).

3. Behavior confirms the two scales differ

Parsed accuracy (parse rate in parentheses):

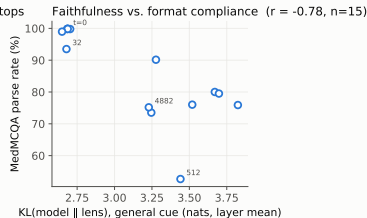
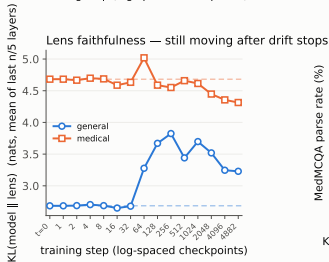
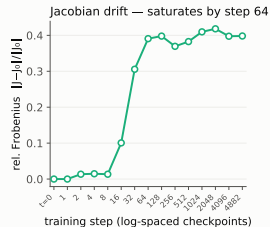
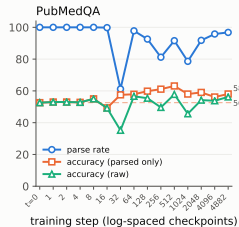
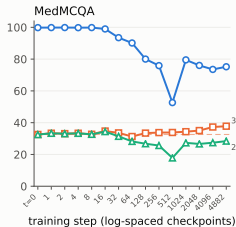
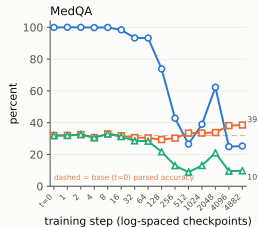
	MedQA		MedMCQA		PubMedQA	
270m-it $t=0$	24.6	(69%)	27.2	(13%)	54.5	(7%)
270m-it final	25.5	(49%)	25.0	(49%)	48.0	(5%)
1b-it $t=0$	31.9	(100%)	32.6	(100%)	52.6	(100%)
1b-it final	38.5	(25%)	37.8	(75%)	58.1	(97%)

- **270m learns nothing.** MedQA/MedMCQA sit near chance throughout; every apparent movement is format churn (MedMCQA parse 13%→49%; PubMedQA swings 5%→88%→5%).
- **1b gains +6.6 / +5.2 / +5.5 pp** parsed — while format compliance collapses, worst on MedQA (100% → 25%).

Reported parsed, not raw: the raw MedQA number would show a *loss* purely from the parse-rate collapse.

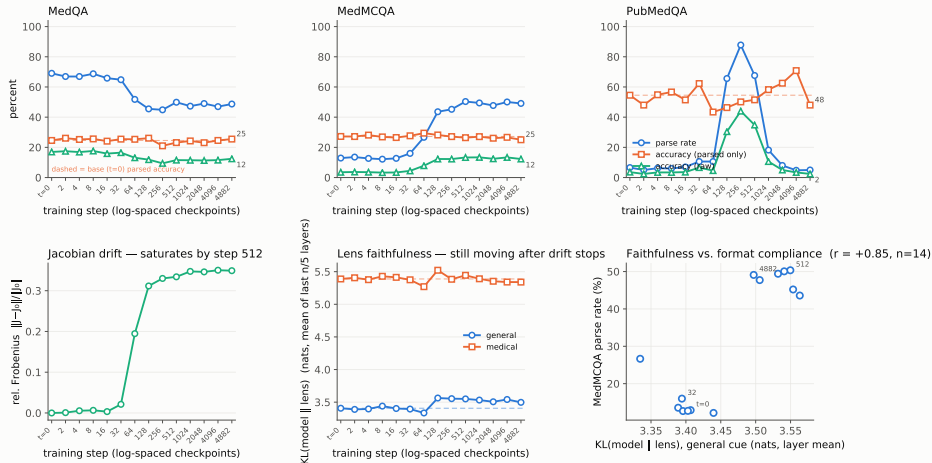
3. Joined behavioral + lens trajectory — 1b-it

Gemma-3-1b-it · medical SFT · behavioral trajectory vs. Jacobian lens



3. Joined behavioral + lens trajectory — 270m-it

Gemma-3-270m-it · medical SFT · behavioral trajectory vs. Jacobian lens



The KL panel is the layer mean, as everywhere else in this deck; at the single final layer the medical curve rises instead.

4. Timing: drift leads format collapse, both lead accuracy

For 1b-it the three curves are offset:

Signal	Onset	Notes
Jacobian drift	steps 8–32	plateaus by step 64
Format compliance	step 32	MedQA parse 100 $\rightarrow$ 93 $\rightarrow$ 74% by 128
Task accuracy	steps 2048+	MedQA still $\sim$ 30 at step 512; 38 only at 4096–4882

Interpretation

The Jacobian has **finished moving before the model has learned any medicine**. Whatever the lens is tracking here is the format / style / alignment shift — not task competence.

Caveats

- **Top-1 is coarse.** $n=50$ prompts for general/medical (93 for multihop), so *per-layer* top-1 quantizes at 0.02; the reported statistic averages 3 layers (270m) or 5 (1b), giving a $1/150$ – $1/250$ grain that makes small moves look more precise than they are. Single-checkpoint wiggles of ± 0.04 are still noise. *Trust the KL trends, not top-1.*
- **Read-out choice is not innocent.** Layer-mean and final-layer KL disagree in sign on 270m medical. Every claim here uses the layer mean; an earlier draft of this deck used the final layer and reported a $270m \leftrightarrow 1b$ *inversion* that does not survive. Report the statistic, not just the number.
- **Two sizes is not a scaling trend.** “1b moves far, 270m barely moves” rests on exactly one pair, and 270m is small enough that it may simply be too weak to learn the task at all. Needs 4b.
- **The general/medical cues are hand-authored,** not sampled from a corpus — the domain-gap result inherits whatever bias is in those 100 prompts. multihop is the one cue with an external, independently-constructed source.
- **Coverage gaps.** `4b_it_validate{,2}` are 5-point wrapper-validation spikes (base + steps 1–8), not trajectories. No -pt arm exists at any size — the pt-vs-it starting-point axis is entirely unstarted.

Next

1. **4b-it full trajectory** — the decisive test of whether “1b moves far, 270m barely moves” is a scale effect or a property of one arm, and whether the layer-mean/final-layer disagreement recurs.
2. **Widen the general/medical cues** beyond the 100 hand-authored prompts, to check the domain-gap collapse is not an artifact of how those pairs were written.
3. **Open a -pt arm** at 1b: does a non-aligned starting point show the same drift-before-competence ordering?
4. **Anchor against medgemma-27b-text-it** on the same probe set to place the trajectory endpoints on an absolute scale.

Data provenance

Artifact	Path
Lens metrics (270m)	eval/jlens/270m_it_full/metrics.jsonl (14 rows)
Lens metrics (1b)	eval/jlens/1b_it_full_v3/metrics.jsonl (15 rows)
Joined traj. (270m)	eval/traj/270m_it_full/joined.json
Joined traj. (1b)	eval/traj/1b_it_full_v3/joined.json
Fit corpus	data/jlens/fit_corpus_v2.txt (1000 WikiText seq, frozen)
Probe set	data/jlens/probe_set_v2.tsv (50/50/93, frozen)
Figures	figs/{270m_it_full,1b_it_full_v3}/
Probe jobs	25776556 (1b), 25799677 (270m)

Both runs logged `fit_corpus=1000 probes=193` — i.e. the current frozen v2 inputs. The v1 files (`fit_corpus.txt`, 40 seq; `probe_set.tsv`, 30 prompts) are legacy and still present in `data/jlens/`; they were used only by the superseded runs.

Superseded: `1b_it_full_v2` (v1 30-prompt set, no multihop), `1b_it_full_probev2` (v2 inputs, immediately re-run as v3), `270m_verify` (Phase-0 8-point smoke test, v1 inputs).